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Polymath Jr

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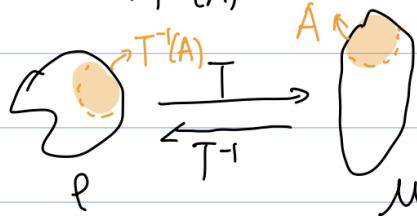
- Optimal Transport problem (Monge Formulation)

Given two densities of mass (probability measures), $\rho, \mu \geq 0$ on $\mathbb{R}^d$

$$\int \rho(x) dx = \int \mu(y) dy = 1,$$

find a map $T: \mathbb{R}^d \rightarrow \mathbb{R}^d$ s.t. $T\# \rho = \mu$ (push forward constraint)

ie. $\int_A \mu(y) dy = \int_{T^{-1}(A)} \rho(x) dx$ for any Borel subset $A \subset \mathbb{R}^d$



and minimizing transportation cost

$$M(T) := \int_{\mathbb{R}^d} c(x, T(x)) \rho(x) dx$$

where $c(x, y)$ is some "cost"

-Eg. $c(x, y) = \frac{1}{2} \|x - y\|^2$

Pushforward Constraint vs. Change of Variables

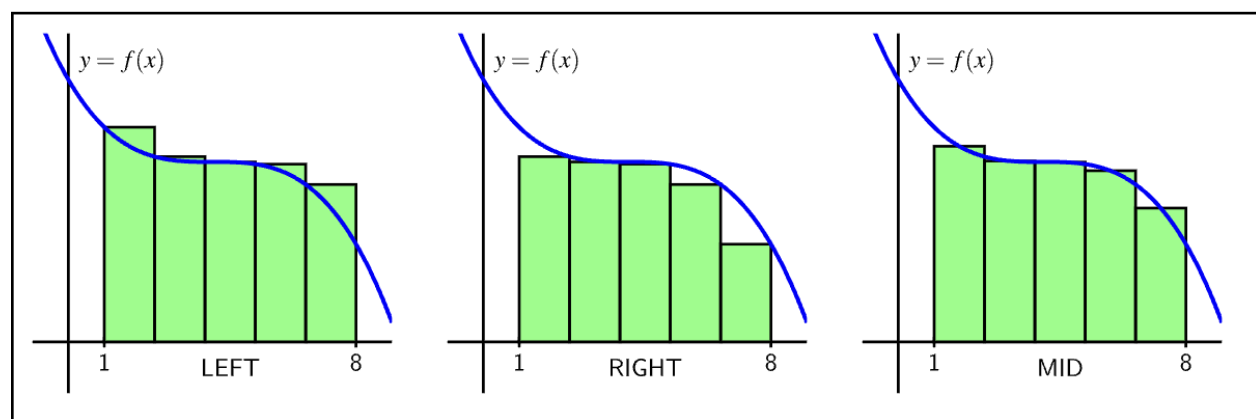
In the Monge formulation of the optimal transport problem, the goal is to find a measurable map $T: \mathbb{R}^d \rightarrow \mathbb{R}^d$ that pushes a source density $\rho(x)$ forward to a target density $\mu(y)$, denoted by $T\# \rho = \mu$. The pushforward condition is expressed as:

$$\int_A \mu(y) dy = \int_{T^{-1}(A)} \rho(x) dx \quad \text{for all Borel sets } A \subset \mathbb{R}^d.$$

This condition guarantees that the total mass assigned to any region A in the target space by μ equals the mass transported into A from the source space under the map T . Importantly, this is a measure-theoretic identity that does *not* involve a change of variables, and thus does *not* require the use of a Jacobian determinant. Both integrals are performed over their respective domains using their native coordinates (dx on the source side and dy on the target side).

Note: In optimal transport (and probability theory in general), requiring conditions to hold for all Borel subsets ensures that we are only dealing with sets for which the integrals are well-defined.

- Density Estimation (using push forward constraint)
 - measure ρ on X , measure μ on Y
 - $T\#\rho \Rightarrow \mu(A) = (T\#\rho)(A) = \rho(T^{-1}(A))$
 - require $T\#\rho = \mu$: $\mathbb{E}_{T\#\rho}[\phi] = \mathbb{E}_{\rho}[\phi \circ T] = \mathbb{E}_{\mu}[\phi]$, $\forall$ measurable func. ϕ

$$\int (\phi \circ T) \rho(x) dx = \int \phi(y) \mu(y) dy$$


Continuous Probability and Density

In the **continuous case**, a single point $X = x$ has **zero probability**, so instead, we describe probability over **intervals** using a **probability density function (PDF)** $f(x)$:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Thus, instead of assigning a probability to a **single point**, we assign probability to a **small interval** $[x, x + \Delta x]$:

$$P(x \leq X \leq x + \Delta x) \approx f(x) \Delta x$$

This means that **for very small** Δx , the discrete probability mass at x corresponds to the density times the interval width:

$$p(x) \approx f(x) \Delta x$$

This is how we bridge our discrete understanding of expectation with the continuous one.

In the discrete case:

$$\mathbb{E}[X] = \sum_x x \cdot p(x)$$

In the continuous case:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

Pushforward and Expectation — Integral Formulation

From the image:

$$T_{\#}\rho = \mu \iff \mathbb{E}_{T_{\#}\rho}[\varphi] = \mathbb{E}_{\rho}[\varphi \circ T] = \mathbb{E}_{\mu}[\varphi] \quad \forall \text{ measurable } \varphi$$

Fully Expanded Integral Form:

$$\int_{\mathbb{R}^d} \varphi(y) d(T_{\#}\rho)(y) = \int_{\mathbb{R}^d} \varphi(T(x)) d\rho(x) = \int_{\mathbb{R}^d} \varphi(y) d\mu(y) \quad \forall \varphi$$

If ρ and μ have densities (with respect to Lebesgue measure), say $\rho(x) dx$ and $\mu(y) dy$, then:

$$\int_{\mathbb{R}^d} \varphi(y) \cdot \underbrace{(T_{\#}\rho)(y)}_{\text{pushforward density} = \mu(y)} dy = \int_{\mathbb{R}^d} \varphi(T(x)) \cdot \rho(x) dx = \int_{\mathbb{R}^d} \varphi(y) \cdot \mu(y) dy$$

Here, $(T_{\#}\rho)(y)$ denotes the density of the pushforward measure, i.e., the amount of mass that ρ transports to y under the map T . If the pushforward condition $T_{\#}\rho = \mu$ holds, then:

$$(T_{\#}\rho)(y) = \mu(y) \quad \text{almost everywhere.}$$

Interpreting the Measure Differential $d\mu(x)$

To interpret $d\mu(x)$ concretely, consider the classical approximation for a continuous random variable X with density $f(x)$:

$$\mathbb{P}(x \leq X \leq x + \Delta x) \approx f(x) \cdot \Delta x$$

This expression tells us that the probability of landing near x is approximately the density at x times the width of the interval.

If μ is the probability measure corresponding to X , then:

$$d\mu(x) = f(x) dx$$

So you can think of $d\mu(x)$ as an infinitesimal piece of probability mass — the analog of $\mathbb{P}(x \leq X \leq x + dx)$. In that sense, when we write:

$$\int_{\mathbb{R}^d} \varphi(x) d\mu(x)$$

we are summing the values of $\varphi(x)$ weighted by the infinitesimal probability mass at each point x . This is the continuous counterpart of a weighted sum like:

$$\sum_i \varphi(x_i) \cdot \mathbb{P}(X = x_i)$$

in the discrete case.

$$\mathbb{E}(g(x)) = \sum_{x: p(x) > 0} g(x) p(x)$$

$$\int \phi(T(x)) p(x) dx = \int \phi(y) \mu(y) dy$$

Change of variable ($y = T(x)$)

$$= \int \phi(T(x)) \mu(T(x)) |T'(x)| dx$$



abs val of det of Jacobian matrix $T(x)$

$$\bullet \quad p(x) = \mu(T(x)) |T'(x)|$$

$$y = T(x)$$

$$x = T^{-1}(y)$$

$$\Rightarrow p(T^{-1}(y)) = \mu(y) \frac{1}{|T^{-1}'(y)|}$$

Jacobian of T^{-1}

Jacobian matrix:

$$x = [x_1, \dots, x_d]^T \in \mathbb{R}^d$$

$$T: \mathbb{R}^d \rightarrow \mathbb{R}^d = \begin{pmatrix} T_1(x_1, \dots, x_d) \\ \vdots \\ T_d(x_1, \dots, x_d) \end{pmatrix}$$

$$T'(x) = \begin{bmatrix} \frac{\partial T_1}{\partial x_1} & \frac{\partial T_1}{\partial x_2} & \dots & \frac{\partial T_1}{\partial x_d} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial T_d}{\partial x_1} & \dots & \dots & \frac{\partial T_d}{\partial x_d} \end{bmatrix}$$

Change-of-Variables Formula and the Jacobian Inverse Identity

We begin with the forward-mode change-of-variables formula. Let $T: \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a bijective, differentiable transformation with differentiable inverse T^{-1} . Suppose that a random variable $x \sim \rho(x)$, and let $y = T(x)$. Then the pushforward density $\mu(y)$ satisfies:

$$\rho(x) = \mu(T(x)) \cdot \left| \det \left(\frac{\partial T}{\partial x}(x) \right) \right|. \quad (1)$$

We now change variables: set $y = T(x)$, so $x = T^{-1}(y)$. Substituting this into equation (1) gives:

$$\rho(T^{-1}(y)) = \mu(y) \cdot \left| \det \left(\frac{\partial T}{\partial x}(T^{-1}(y)) \right) \right|. \quad (2)$$

Using the **Jacobian inverse identity**, we have:

$$\left| \det \left(\frac{\partial T}{\partial x}(T^{-1}(y)) \right) \right| = \frac{1}{\left| \det \left(\frac{\partial T^{-1}}{\partial y}(y) \right) \right|}. \quad (3)$$

Substituting this into equation (2) yields the alternative form of the change-of-variables formula:

$$\rho(T^{-1}(y)) = \mu(y) \cdot \frac{1}{\left| \det \left(\frac{\partial T^{-1}}{\partial y}(y) \right) \right|}. \quad (4)$$

• Monte Carlo Integration identically independent distributed
↓

a R.V. $X \sim \rho(x)$ PDF (density), $X_1, \dots, X_N$ iid samples $\sim \rho$

• $\mathbb{E}[\varnothing] = \mathbb{E}_\rho[\varnothing(x)] = \int \varnothing(x) \rho(x) dx$

$\approx \frac{1}{N} \sum_{i=1}^N \varnothing(x_i)$ (sample average)

(CLT, Law of Large # $\Rightarrow$ convergence theorem)

Weak Law of Large Numbers (WLLN)

If $X_1, \dots, X_n \sim \rho(x)$ are i.i.d. and $\Theta(X_i)$ has finite mean μ , then the empirical average:

$$A_n = \frac{1}{n} \sum_{i=1}^n \Theta(X_i)$$

converges in probability to μ :

$$\forall \varepsilon > 0, \quad \mathbb{P}(|A_n - \mu| > \varepsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

This guarantees that if you draw more and more samples, your Monte Carlo estimate will be close to the true expectation **most of the time**.

This is what allows you to approximate integrals using random sampling!

Central Limit Theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with finite mean μ and variance α^2 . Define:

$$S_n = \sum_{i=1}^n X_i, \quad \text{and} \quad B_n = \frac{S_n - n\mu}{\sqrt{n\alpha^2}}$$

Then the Central Limit Theorem states:

$$B_n \xrightarrow{\text{law}} \mathcal{N}(0, 1)$$

Step-by-Step Breakdown

Step 1: The Setup

You start with i.i.d. variables:

- $X_i \sim \rho(x)$
- Each has finite mean $\mu = \mathbb{E}[X_i]$
- Each has finite variance $\alpha^2 = \text{Var}(X_i)$

Define the sum:

$$S_n := X_1 + X_2 + \cdots + X_n \quad \Rightarrow \quad \mathbb{E}[S_n] = n\mu, \quad \text{Var}(S_n) = n\alpha^2$$

So the expected total is $n\mu$, and the standard deviation is $\sqrt{n\alpha^2}$.

Why Does $\text{Var}(S_n) = n\alpha^2$?

We define the sum:

$$S_n = X_1 + X_2 + \cdots + X_n$$

where each $X_i \sim \rho(x)$ are i.i.d. random variables with variance $\text{Var}(X_i) = \alpha^2$.

Important Note: Variance is **not a linear operator** like expectation. In general,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j)$$

So unless the random variables are uncorrelated (i.e. $\text{Cov}(X, Y) = 0$), we cannot simply add their variances.

Key Fact: If $X_1, \dots, X_n$ are **independent**, then

$$\text{Cov}(X_i, X_j) = 0 \quad \text{for all } i \neq j$$

which implies:

$$\text{Var}(S_n) = \sum_{i=1}^n \text{Var}(X_i) = n\alpha^2$$

The identity $\text{Var}(S_n) = n\alpha^2$ holds **only because the variables are independent**. This is what allows us to scale variance linearly in this case.

Step 2: Center and Scale

We define:

$$B_n := \frac{S_n - n\mu}{\sqrt{n\alpha^2}}$$

This means:

- You center S_n by subtracting its mean $n\mu$
- You scale it to unit variance by dividing by $\sqrt{n\alpha^2}$

Step 3: The Conclusion

The CLT tells us:

$$B_n \xrightarrow{\text{law}} \mathcal{N}(0, 1)$$

That is, for large n , the distribution of B_n approaches a standard normal. So even if the original X_i are not normal, the sum becomes approximately normal after proper centering and scaling.

Why This Matters for Monte Carlo

We begin with the Central Limit Theorem normalization:

$$B_n := \frac{S_n - n\mu}{\sqrt{n\alpha^2}}$$

Now recall that the empirical (Monte Carlo) average is:

$$A_n = \frac{1}{n} \sum_{i=1}^n X_i = \frac{S_n}{n} \quad \Rightarrow \quad S_n = nA_n$$

Substitute this into the expression for B_n :

$$B_n = \frac{nA_n - n\mu}{\sqrt{n\alpha^2}} = \frac{n(A_n - \mu)}{\sqrt{n\alpha^2}} = \frac{A_n - \mu}{\alpha/\sqrt{n}}$$

So we can express B_n directly in terms of A_n :

$$B_n = \frac{A_n - \mu}{\alpha/\sqrt{n}}$$

Interpretation

This shows that for large n , the Monte Carlo estimate A_n satisfies:

$$A_n \approx \mu + \mathcal{N}\left(0, \frac{\alpha^2}{n}\right)$$

To see why, start from the Central Limit Theorem statement:

$$B_n = \frac{A_n - \mu}{\alpha/\sqrt{n}} \xrightarrow{\text{law}} \mathcal{N}(0, 1)$$

Multiply both sides by $\alpha/\sqrt{n}$:

$$A_n - \mu \xrightarrow{\text{law}} \mathcal{N}\left(0, \frac{\alpha^2}{n}\right)$$

Then add μ to both sides:

$$A_n \xrightarrow{\text{law}} \mu + \mathcal{N}\left(0, \frac{\alpha^2}{n}\right)$$

This tells us that the Monte Carlo average A_n converges in distribution to a normal distribution centered at the true mean μ , with variance shrinking as $1/n$.

- It stays close to μ , as guaranteed by the Law of Large Numbers.
- The error follows an approximately Gaussian distribution.
- The standard deviation of the noise decays like $\frac{1}{\sqrt{n}}$, meaning more samples reduce uncertainty.

$$\mathbb{E}_\rho[\Theta(x)] = \int \Theta(x) \frac{\rho(x)}{\tilde{\rho}(x)} \cdot \tilde{\rho}(x) dx = \mathbb{E}_{\tilde{\rho}}[\Theta(x) \cdot \frac{\rho}{\tilde{\rho}}(x)]$$

$$\text{(weighted sum)} \approx \frac{1}{N} \sum_{i=1}^N \Theta(x_i) \cdot \frac{\rho}{\tilde{\rho}}(x_i)$$

weight

Importance Sampling: Weighted Monte Carlo Estimation

Suppose we want to compute the expectation of a function $\Theta(x)$ under a probability density $\rho(x)$:

$$\mathbb{E}_\rho[\Theta(X)] = \int \Theta(x) \rho(x) dx$$

However, sampling from $\rho(x)$ may be difficult. Instead, assume we can sample from an easier distribution $\tilde{\rho}(x)$, called the *proposal* or *reference* distribution.

We rewrite the expectation by multiplying and dividing by $\tilde{\rho}(x)$:

$$\mathbb{E}_\rho[\Theta(X)] = \int \Theta(x) \frac{\rho(x)}{\tilde{\rho}(x)} \tilde{\rho}(x) dx = \mathbb{E}_{\tilde{\rho}} \left[\Theta(x) \cdot \frac{\rho(x)}{\tilde{\rho}(x)} \right]$$

Here, the correction factor

$$w(x) := \frac{\rho(x)}{\tilde{\rho}(x)}$$

is called the *importance weight*. It corrects for the fact that we're using samples from the wrong distribution.

Monte Carlo Approximation

If $x_1, \dots, x_N \sim \tilde{\rho}(x)$ are i.i.d. samples from the reference distribution, then:

$$\mathbb{E}_\rho[\Theta(X)] \approx \frac{1}{N} \sum_{i=1}^N \Theta(x_i) \cdot \frac{\rho(x_i)}{\tilde{\rho}(x_i)}$$

This is called an **importance sampling estimate**. It is a weighted sum of the function values $\Theta(x_i)$, where each term is scaled by its corresponding importance weight $w(x_i)$.

Interpretation

- When $\tilde{\rho}(x)$ is similar to $\rho(x)$, the importance weights

$$w(x) = \frac{\rho(x)}{\tilde{\rho}(x)}$$

are close to 1 for most samples. This means you're sampling from the right regions with the right frequency, and the estimate is stable.

- If $\tilde{\rho}(x)$ assigns very little probability to regions where $\rho(x)$ is large (i.e., it under-samples important areas), then the weights $w(x)$ become large to compensate. This ensures those rare but important samples contribute more — but it can also introduce high variance.
- If the weights vary too much — some being extremely large while others are close to zero — the resulting estimate can become unreliable. A few outlier samples may dominate the weighted sum, making the approximation noisy or unstable.

Bottom line: For importance sampling to work well, the proposal distribution $\tilde{\rho}(x)$ must be carefully chosen to cover the regions where $\rho(x) \cdot \Theta(x)$ is large. A poor choice leads to high-variance or biased estimates.

• Gradient Flow / Gradient Descent

• ODE

$$\begin{array}{c} \swarrow \text{time} \\ x(t) \\ \uparrow \\ \text{position/displacement} \end{array} \quad (\text{velocity}) \dot{x} = \frac{dx}{dt} = V(x(t))$$

$$x(t) = x(t_0) + \int_{t_0}^t V(x(s)) ds$$

$$\text{— eg. } V(x) = V \text{ const.} \Rightarrow x(t) = x(t_0) + V(t - t_0)$$

• Gradient Flow

$$\left(\frac{x(t_{n+1}) - x(t_n)}{\Delta t} \approx \right) \dot{x} = \frac{dx}{dt} = -\nabla_x F(x(t)) = -\nabla_x \overset{\text{some objective to minimize}}{F}(x)$$

The particle stops moving exactly when it reaches a critical point of F , where the gradient is zero.

$$\Rightarrow \dot{x} = 0 \text{ (velocity} = 0) \Rightarrow \nabla_x F = 0$$

Understanding Gradient Flow

Gradient descent is a discrete-time optimization algorithm that updates parameters using the rule:

$$x_{t+1} = x_t - \eta \nabla F(x_t),$$

where $\eta > 0$ is the step size (learning rate), and $\nabla F(x_t)$ is the gradient of the objective function F evaluated at x_t . At each time step, the algorithm takes a finite step in the direction of steepest descent.

Gradient flow is the continuous-time analogue of gradient descent. It describes a continuous trajectory $x(t)$ governed by the differential equation:

$$\frac{dx}{dt} = -\nabla F(x(t)).$$

This equation defines a dynamical system in which the parameter vector $x(t)$ evolves smoothly over time, always moving in the direction of steepest decrease of the objective function.

The connection between gradient descent and gradient flow arises by taking the limit as the step size $\Delta t \rightarrow 0$. In this limit, the finite difference update:

$$\frac{x(t + \Delta t) - x(t)}{\Delta t} \approx -\nabla F(x(t))$$

becomes the continuous-time derivative $\frac{dx}{dt}$, leading to the gradient flow equation.

Gradient flow provides a mathematically precise framework for analyzing optimization dynamics. Unlike discrete updates, which occur at isolated time points, gradient flow describes a continuous trajectory along which the objective function decreases. This formulation allows for the application of tools from differential equations to study convergence, stability, and critical point behavior.

In practice, gradient descent is an approximation of gradient flow. As the step size becomes small, the discrete updates more closely follow the continuous trajectory defined by gradient flow. While gradient descent is used for numerical optimization, gradient flow is typically used for theoretical analysis and understanding the qualitative behavior of learning dynamics.

• Normalizing Flow

$$\frac{dT(x, t)}{dt} = - \frac{\delta \text{KL}(\phi \circ T(\cdot, t))}{\delta \phi} \Big|_{\phi = \text{id}}$$

Gradient Flow: KL Minimization in Function Space

The expression

$$\frac{dT(x, t)}{dt}$$

represents the change in the transformation T with respect to time t , while holding the input x fixed.

Why the Minus Sign?

The gradient flow equation is:

$$\frac{dT(x, t)}{dt} = - \frac{\delta}{\delta \phi} \text{KL}[\phi \circ T(\cdot, t)] \Big|_{\phi = \text{id}}$$

This is not the full gradient of KL in function space. It's a directional derivative: it tells you, for a given direction ϕ , how much you should move in that direction to increase KL most steeply along that path.

This is just like how $\frac{\partial}{\partial x}(x^2 + y^2) = 2x$ doesn't tell you how much the function increases — it tells you how steep the function is along x , and how strongly you should move in the x -direction if you want to increase f fastest with respect to x .

So here, the variational derivative with respect to ϕ tells you how much to move in that direction to increase KL steeply. To minimize KL, we go in the opposite direction — hence the minus sign.

Why Use x on the LHS and $\cdot$ on the RHS?

In the expression

$$\frac{dT(x, t)}{dt},$$

we're taking the partial derivative of the function $T(x, t)$ with respect to time t , treating x as a constant symbol during the operation. We are not evaluating the derivative at a specific value of x , and it does not matter what x is. That's how partial derivatives work — other variables are just treated as frozen symbols, not plugged-in values.

On the right-hand side:

$$\left. \frac{\delta}{\delta \phi} KL[\phi \circ T(\cdot, t)] \right|_{\phi=\text{id}},$$

we write $T(\cdot, t)$ using a dot to emphasize that we are viewing T as a full function — not as something evaluated at a single point. This is because the KL divergence is an integral over all x , which we evaluate with the Monte Carlo estimation.

Why We Drop $\rho(x)$ in KL Divergence Approximation

We begin with the KL divergence between a true (unknown) data distribution $\rho(x)$ and a model-induced density $\tilde{\rho}(x)$:

$$KL[\rho \parallel \tilde{\rho}] = \int \rho(x) \log \left(\frac{\rho(x)}{\tilde{\rho}(x)} \right) dx$$

In our setting, the model density $\tilde{\rho}(x)$ is defined as the pushforward of a base density μ through a map T_β , using the change-of-variables formula:

$$\tilde{\rho}(x) = \mu(T_\beta(x)) \cdot \left| \det \left(\frac{\partial T_\beta}{\partial x}(x) \right) \right|$$

Substituting into the KL divergence, we get:

$$KL[T_\beta] = \int \rho(x) \log \left(\frac{\rho(x)}{\mu(T_\beta(x)) \cdot \left| \det \left(\frac{\partial T_\beta}{\partial x}(x) \right) \right|} \right) dx$$

If we instead used $\phi \circ T_\beta$, the KL divergence becomes:

$$KL[\phi \circ T_\beta] = \int \rho(x) \log \left(\frac{\rho(x)}{\mu(\phi(T_\beta(x))) \cdot \left| \det \left(\frac{\partial (\phi \circ T_\beta)}{\partial x}(x) \right) \right|} \right) dx$$

This expression shows how the pushforward density changes when an additional transformation ϕ is composed on top of T_β , which is necessary to know when taking variational derivatives with respect to ϕ .

$$T(\cdot, t^n) = \phi^n \circ \phi^{n-1} \circ \dots \circ \phi^1$$

$$\phi^k = \phi(\cdot, \theta^k) \quad , \quad \text{assume } \phi(\cdot, \theta_0) = \text{id}.$$

Understanding $\phi(\cdot, \theta)$ Notation

In the context of normalizing flows or parametric transformations, we often encounter expressions like $\phi(\cdot, \theta)$. This notation reflects that ϕ is a function of two inputs:

- an input point $x \in \mathbb{R}^d$,
- and a set of parameters $\theta \in \mathbb{R}^p$.

The output is a transformed point $\phi(x, \theta) \in \mathbb{R}^d$. When we write $\phi(\cdot, \theta^k)$, we mean that the parameters θ^k are fixed, and we view the function $x \mapsto \phi(x, \theta^k)$ as a transformation applied to the input space.

Composition of Maps: Step-by-Step Action

Suppose we begin with a sample $x_0 \sim \rho$, where ρ is the target data distribution. The transformation is built by composing several parameterized functions $\phi^k = \phi(\cdot, \theta^k)$, where each $\phi(x, \theta)$ takes:

- an input point $x \in \mathbb{R}^d$,
- and parameters $\theta \in \mathbb{R}^p$,

and returns:

- the transformed point $\phi(x, \theta) \in \mathbb{R}^d$.

The data flows through these maps as follows:

$$\begin{aligned} x_1 &= \phi^0(x_0) \\ x_2 &= \phi^1(x_1) = \phi^1(\phi^0(x_0)) \\ x_3 &= \phi^2(x_2) = \phi^2(\phi^1(\phi^0(x_0))) \\ &\vdots \end{aligned}$$

Each transformation ϕ^k applies to the output of the previous transformation. This chained composition produces a final transport map $T = \phi^n \circ \dots \circ \phi^0$, which pushes samples from ρ toward a base distribution μ .

$$KL(T(\cdot, t)) = \int \log \left(\frac{p(x)}{\tilde{p}_t(x)} \right) p(x) dx \quad (\text{"difference btw } p \text{ \& } \tilde{p}_t")$$

$$\rightarrow T(\cdot, t) \# \rho \sim \tilde{\rho}_t$$

$$KL_t = KL(\phi \circ T(\cdot, t))$$

$$\tilde{\rho}_t^\phi \sim \phi \circ T(\cdot, t) \# \rho$$

The symbol after the # is always the source distribution — the one from which you are sampling before applying the transformation

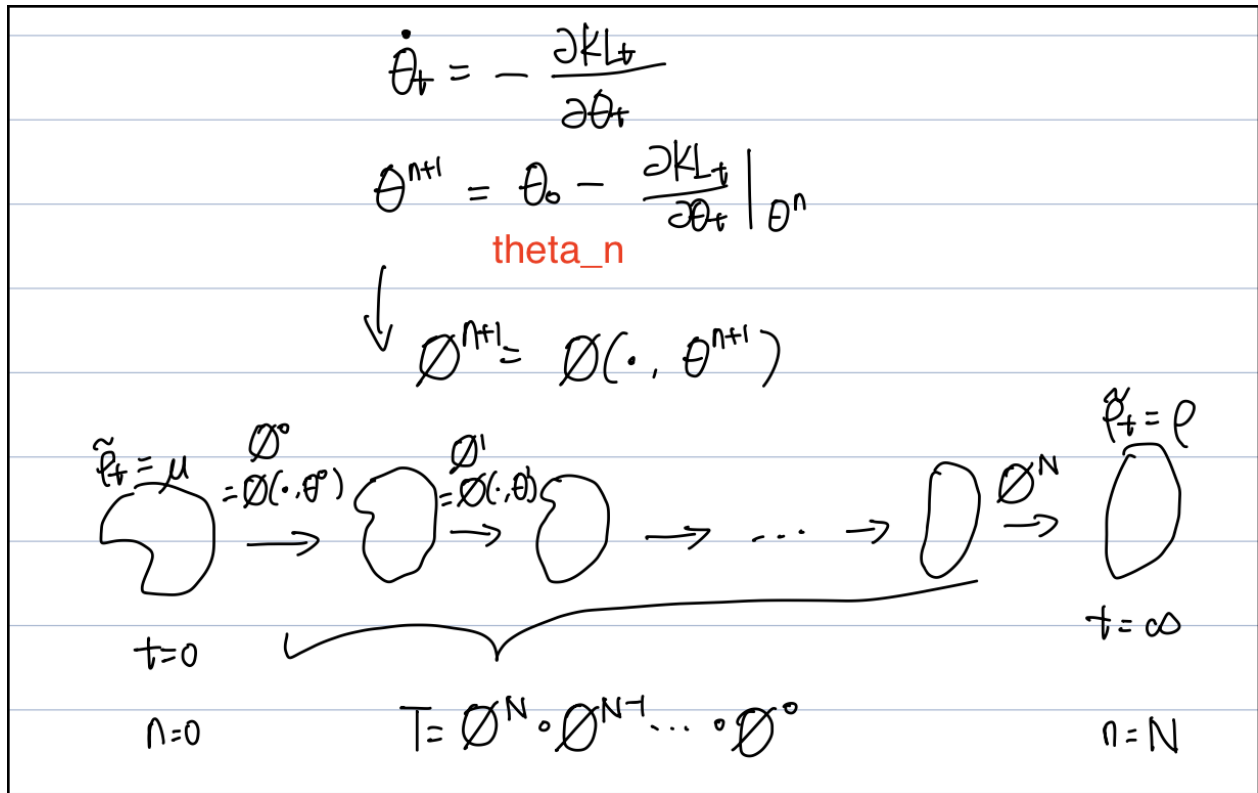
$$\phi(T(\cdot, t), \theta(t)) \sim \tilde{\rho}_t^\phi$$

Meaning of the Expression $\phi(T(\cdot, t), \theta(t)) \sim \tilde{\rho}_t^\phi$

This expression means:

- Take the map $T(\cdot, t)$, which evolves over time t ,
- Apply another transformation $\phi(\cdot, \theta(t))$ to it — where ϕ is parameterized by $\theta(t)$,

Then the resulting composition has distribution $\tilde{\rho}_t^\phi$. That is, the output of $\phi \circ T$ (acting on samples from ρ for input) is distributed according to $\tilde{\rho}_t^\phi$.



We begin with a target distribution ρ , from which we sample directly, and construct a sequence of transformations $\varphi^0, \varphi^1, \dots, \varphi^N$, each parameterized by θ^n , to map these samples progressively toward an evolving approximation $\tilde{\rho}_t$ of a simpler base distribution μ . The transformations are composed as

$$T = \varphi^N \circ \varphi^{N-1} \circ \dots \circ \varphi^0,$$

and at each stage, we update the parameters by minimizing the Kullback–Leibler divergence

$$\text{KL}(\varphi^n \circ \dots \circ \varphi^0 \# \rho \parallel \rho).$$

This construction ensures that the pushforward $\tilde{\rho}_t := \varphi^n \circ \dots \circ \varphi^0 \# \rho$ increasingly resembles the base measure ρ as $t \rightarrow \infty$.